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$$K_p = \frac{x^2}{(a-x)V}\frac{PV}{n}$$

$$K_p = \frac{x^2}{(a-x)V}\frac{PV}{(a+x)}$$

$$K_p = \frac{x^2P}{(a-x)(a+x)}$$

Synthesis of ammonia:

Let us consider the formation of ammonia in which, 'a' moles nitrogen and 'b' moles hydrogen gas are allowed to react in a container of volume V. Let 'x' moles of nitrogen react with 3x moles of hydrogen to give 2x moles of ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

	N_2	H_2	NH_3
Initial number of moles	a	b	0
number of moles reacted	x	3x	0
Number of moles at equilibrium	a-x	b-3x	2x
Active mass or molar concentration at equilibrium	$\frac{a-x}{V}$	$\frac{b-3x}{V}$	$\frac{2x}{V}$

Applying law of mass action,

$$K_C = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$= \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-3x}{V}\right)^3}$$

$$= \frac{\left(\frac{4x^2}{V^2}\right)}{\left(\frac{a-x}{V}\right)\left(\frac{b-3x}{V}\right)^3}$$

$$K_C = \frac{4x^2V^2}{(a-x)(b-3x)^2}$$

The equilibrium constant K_p can also be calculated as follows:

$$K_p = K_C(RT)^{(\Delta n_g)}$$

$$\Delta n_g = n_p - n_r = 2 - 4 = -2$$

$$K_p = \frac{4x^2V^2}{(a-x)(b-3x)^3}(RT)^{-2}$$

Total number of moles at equilibrium,

n= a-x + b-3x+ 2x = a+b-2x

$$K_p = \frac{4x^2V^2}{(a-x)(b-3x)^3} \times \left[\frac{PV}{n}\right]^{-2}$$

$$K_p = \frac{4x^2V^2}{(a-x)(b-3x)^3} \times \left[\frac{n}{PV}\right]^{2}$$

$$K_p = \frac{4x^2V^2}{(a-x)(b-3x)^3} \times \left[\frac{a+b-2x}{PV}\right]^{2}$$

$$K_p = \frac{4x^2(a+b-2x)^2}{P^2(a-x)(b-3x)^3}$$

Solved Problems:

1. The equilibrium concentrations of NH_3, N_2 and H_2 are 1.8×10^{-2} M, 1.2×10^{-2} M and 3×10^{-2}M respectively. Calculate the equilibrium constant for the formation of NH_3 from N_2 and H_2. [Hint: M= mol lit^{-1}]